BACHELOR OF VOCATION

(AGRI-BUSINESS AND AGRARIAN ENTREPRENEURSHIP)

Semester I								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
*GEN -101	Communication Skills	Generic Component	Theory	20	80	----	----	6
*GEN-102	Fundamentals of Information Technology	Generic Component	Theory	20	80	----	----	6
ASC -103	Dairy Farm Management	Skill Component	Theory & Practical	10	40	10	40	6
ASC -104	Live Stock Management - I	Skill Component	Theory & Practical	10	40	10	40	6
ASC -105	Feed and Water Management	Skill Component	Theory & Practical	10	40	10	40	6
Semester II								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
*GEN -201	Soft Skills and Personality Development	Generic Component	Theory	20	80	----	----	6
GC 202	Business Ethics	Generic Component	Theory	20	80	--	--	6
ASC-203	Fodder and Forage Management	Skill Component	Theory & Practical	10	40	10	40	6
ASC-204	Live Stock Management - II	Skill Component	Theory & Practical	10	40	10	40	6
ASC-205	Workforce Health & Safety Management	Skill Component	Theory & Practical	10	40	10	40	6

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

Semester III								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
*GEN -301	Value Education And Human Rights	Generic Component	Theory	20	80	----	----	6
GC 302	E-Commerce	Generic Component	Theory	20	80	----	----	6
ASC -303	Introduction to Poultry Farming	Skill Component	Theory & Practical	10	40	10	40	6
ASC -304	Management of Broiler & Layer Chicken	Skill Component	Theory & Practical	10	40	10	40	6
ASC -305	Hatchery Management	Skill Component	Theory & Practical	10	40	10	40	6
Semester IV								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
*GEN 401	Environmental Studies	Generic Component	Theory	20	80	----	----	6
GC 402	Project Management	Generic Component	Theory	20	80	--	--	6
ASC-403	Poultry Nutrition	Skill Component	Theory & Practical	10	40	10	40	6
ASC-404	Poultry Health & Bio Security	Skill Component	Theory & Practical	10	40	10	40	6
ASC-405	Poultry Production Economics & Marketing	Skill Component	Theory & Practical	10	40	10	40	6

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

Semester V								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
*GEN 501	Critical Thinking And Elementary Statistics	Generic Component	Theory	20	80	----	----	6
GC 502	Introduction To Research Methodology And Report Writing	Generic Component	Theory	20	80	----	----	6
ASC -503	Introduction to Shrimp Farming	Skill Component	Theory & Practical	10	40	10	40	6
ASC -504	Seed Stocking and Health Management of Shrimp	Skill Component	Theory & Practical	10	40	10	40	6
ASC -505	Feed and Water Management in Shrimp Farming	Skill Component	Theory & Practical	10	40	10	40	6
Semester VI								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
*GEN 601	Entrepreneurship Development Programme	Generic Component	Theory	20	80	----	----	6
GC 602	Total Quality Management	Generic Component	Theory	20	80	--	--	6
ASC-603	Industrial Training	Skill	Practical				300	18

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

Semester I

Title: ASC- 103 Dairy Farm Management

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: the objective of the course is to provide knowledge about the process of planning and budgeting of a dairy farm, market structure of the dairy industry and the relevance of collection of market information.

Job Role: The Dairy Farmer/Entrepreneur is expected to take various decisions regarding viability and sustainability of dairy farms. He/she is expected to ensure proper care of dairy animals, their health and productivity, milking and marketing of dairy products.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

Economic aspects of Dairy Farming:Basic steps of dairy farm planning and budgeting; Basic principles of keeping books of accounts; Various Government and other schemes / products / offers available for dairydevelopment and milk marketing

Unit II

Market information management:Various methods of updating oneself with market information such as mobileusage, contact with key informants, tie up government agencies etc.

Unit III

Marketing aspects of Dairy Farming: Quality parameters of milk and milk products and their market prices; Pricing mechanism of various buyers of milk and milk products; Costing of various logistic arrangements towards the sale of milk and milkproducts at different markets and consumer points

Unit IV

Different players selling various dairy farm inputs and their prices; Different players buying milk and milk products and their prices. Needs and options available with various clients; Advantages and disadvantages of doing business with each one of the clients.

Practical: To conduct a survey and prepare a report on the cost of starting a dairy business; to assess market structure of dairy business in the local area and in general.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

1. "Handbook of Dairy Farming", EIRI Books
2. Kat Bazeley and Alastair Hayton , "Practical Cattle Farming", The Crowood Press Ltd
3. "Dairy Farm Management", FAO report

Title: ASC - 104 Live Stock Management – I

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: The course is designed to impart knowledge about the preparation of livestock accommodation and its maintenance. Also, the placing of livestock in the accommodation and monitoring the livestock placed.

Job Role: The Dairy Farmer/Entrepreneur is expected to take various decisions regarding viability and sustainability of dairy farms. He/she is expected to ensure proper care of dairy animals, their health and productivity, milking and marketing of dairy products.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit – I

Preparation of Livestock Accommodation: Types of animal accommodation and suitability for different animals; Animals' welfare requirements; Animal welfare legislation and codes of best practice in animal welfare andanimal accommodation; Materials and equipment required in animal accommodation;Containing structures for different animals(indoors and outdoors); Potential hazards in relation to the accommodation; Environmentally sound methods for managing animal waste; Effects of different cleaning methods and materials on thehealth and wellbeing of animals.

Unit –II

Maintenance of Livestock accommodation: Factors taken into account when monitoring including environmentalconditions, containing structures, materials and equipment, weather conditions andthe density and mix of the animals; Monitoring the preparation and maintenance of accommodation, quality ofworking methods and practice, resource use and suitability, scheduling, the health andwell-being of the animals and the cleanliness and suitability of the accommodation; Indicators of animal health and development and signs of ill health; Options available for changing plans for accommodation; Animal welfare legislation and codes of best practice in animal welfare andanimal accommodation; Effective health and safety methodswhile maintaining animal accommodation.

Unit - III

Establish livestock in their accommodation: Appropriate livestock accommodations required for each type of animal; Materials (e.g. bedding) and environmental conditions which animals needwithin their accommodation to maintain their health and well-being; Different factors to be taken into account when preparingaccommodation in a safe, secure and clean state; Importance of settling animals in a new place; Specific needs of special care animals; Methods of handling livestock safely

Unit – IV

Monitor livestock in their accommodation: Relationship between maintaining animals accommodation and promotinganimal health and well-being; Recognize stress, normal and abnormal behaviour within livestock; Specific needs of special care animals; Cleaning routines appropriate to the animal species concerned and relevantaccommodation; Hygiene standards, disinfectants, cleaning agents, cleaning techniques andcleaning equipment and materials; Effects of different cleaning methods and materials on thehealth and well-being of animals; Organizational health and safety and animal welfare policy; Consequences of failing to monitor effectively or not reporting variations

Practical: To prepare notes and draw conclusions from the same about the livestock management practices practiced by the dairy farmers in the neighbourhood.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.

- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

1. Richard A. Battaglia, "Handbook of livestock management" Pearson Education.
2. N.S.R. Sastry and C.K. Thomas, "Livestock Production and management", Kalyani Publishers, Ludhiana, India.
3. V.K. Taneja, C. Chakravarty, C.S. Viswanath, Aruna T. Kumar "Hand book of Animal Husbandry", Indian Council of Agricultural Research, New Delhi.

Title: ASC - 105 Feed and Water Management

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: the objective of the course is to provide knowledge about the feed quality, procurement, storage, feeding process, the provision of water, the equipment required for feeding and watering the livestock.

Job Role: The Dairy Farmer/Entrepreneur is expected to take various decisions regarding viability and sustainability of dairy farms. He/she is expected to ensure proper care of dairy animals, their health and productivity, milking and marketing of dairy products.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.

- All questions carry equal marks

Unit – I

Receiving and Storage of Animal Feed: Checking correct quantities and types of animal feed have been deliveredand that they are in an acceptable condition; Report any shortfalls or faults in delivered animal feed to the appropriate person; Handle animal feed safely, and in a way that protects it from damage and contamination and minimizes wastage; Storage of animal feed safely according to the manufacturer's recommendations, inthe order in which it is to be used; Check stored animal feed regularly for any signs of pest infestation and reportthis to the appropriate person straight away; Check stock levels regularly and tell the appropriate person if stocks are low.

Unit – II

Preparation of feed and water supplies for livestock: Basic nutritional requirements for animals,including water storage, preparation and presentation requirements for feed and feed supplements; Identify the type and quantity of animal feed required for each class of livestock and their basic role in animal diets; Prepare feed for livestock; how to use and store equipment correctly

Unit III

Monitoring and maintenance of supply of feed and water to livestock: Correct method of supplying feed and maintaining adequate levels; Reasons for checking the condition of feed and water; Methods of cleaning and maintaining feeding and watering equipment in a fit condition.

Unit IV

Importance of ensuring all livestock have access to feed and water; Identifying normal feeding and drinking behaviour of livestock and potential reasons for changes in consumption; Correct methods for disposing of organic and inorganic feed waste.

Practical: To prepare notes and draw conclusion from the same regarding the feed and water management practices followed by the dairy farmers in the neighbourhood.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.

- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

1. Keith Wheately , "Agricultural water Management: Insights and Challenges"
2. Roy Kellaway and Tim Harrington. "Feeding Concentrates: Supplements for Dairy Cows"
3. John Morgan and Scott Mc Donald "Feed Pads for Grazing Dairy Cows"

Semester II

Paper Title: GEN 202 - BUSINESS ETHICS

Theory:80
Internal Assessment: 20
Time:3 hours

Objective: The objective of this paper is to familiarize the students with the importance of ethics in business and understanding of issues related to corporate social responsibility and corporate governance.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Unit 1

Business Ethics: Meaning and Concept, Principles of Business Ethics, Characteristics of Ethical Organisations, Theories of Business Ethics.

Unit-II

Globalization and Business Ethics, Stakeholder's Protection, Corporate Governance and Business Ethics. Ethical Issues in Indian Business.

Unit - III

Professional Values: Objectives, Morals, Values, Ethics, Integrity, Work ethics, Service learning, Virtues, Respect for others, Living peacefully, Caring, Sharing, Honesty, Courage, Valuing time, Cooperation, Commitment, Empathy, Self-confidence, Challenges in the work place, Spirituality.

Unit - IV

Corporate Social Responsibility**:** Social Responsibility of business with respect to different stakeholders, Arguments for and against social responsibility of business, Social Audit, Corporate Social Responsibility and Corporate Governance.

Suggested Readings:

1. J.P. Sharma, *Corporate Governance, Business Ethics & CSR,* Ane Books Pvt. Ltd., Nee Delhi.
2. Andrew Crane, Dirk Matten, *Business Ethics,* Oxford University Press, New Delhi.
3. Daniel Albuquerque, *Business Ethics, Principles and Practices (Indian Edition),* Oxford University Press, New Delhi
4. Fr. Floriano C. Roa, *Business Ethics and Social Responsibility*, Rexestore.
5. O. C. Ferrell, John Fraedrich, Linda Ferrell, *Business Ethics: Ethical Decision Making & Cases, Cengagae Learning*
6. Michael Blowfield, Alan Murray, *Corporate Responsibility – A Critical Introduction,* Oxford University Press, New Delhi.

Title: ASC - 203 Fodder and Forage Management

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: the course aims to equip the trainee with the knowledge about the fodder and forage harvesting,the equipment to be used and its maintenance, its storage and preservation and maintenance of safety standards.

Job Role: The Dairy Farmer/Entrepreneur is expected to take various decisions regarding viability and sustainability of dairy farms. He/she is expected to ensure proper care of dairy animals, their health and productivity, milking and marketing of dairy products.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit - I

Fodder Management: Various forage conservation options based on the year roundrequirement of feed and fodder; Risk factors for spoilage in forage conservation such as fire, vermin andair in silage; Favourable and unfavourable conditions for forage conservation; Forage conservation machinery and equipment,

Unit - II

Occupational Health and Safety (OHS) requirements; Access to paddocks for harvesting and transport machinery; Storage facility for selected forage conservation method.Safety hazards and OHS procedures for forage conservation.

Unit- III

Forage Management: Weather conditions to determine optimum time for harvest; Dry matter target and assessing dry matter content of forage material for the forage operation; Mow, condition, tender and/or rake swaths depending on weather conditions and forage drying targets; Conduct of harvesting activities in a safe, controlled and efficient manner;

Unit - IV

Equipment required during harvesting operations; Bale, wrap, compact, seal or store forage in accordance with storage plan; Load, transport, and store/compact (if required) forage in accordance with OHSand quality requirements; Store forage so as to minimise risk of spoilage and combustion; Disposal of waste and debris; Test or sample stored forage for quality; Mitigate environmental impacts of forage conservation activities.

Practical: A detailed project is to be undertaken on the fodder and forage management. The equipment used in harvesting and techniques used for storage and preservation needs are to be highlighted. Measures for quality control are to be assessed.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

1. Forages and fodder by Anil Kumar Singh, M A Khan, NatrajaSubash and Krishna Murari Singh.
2. Forages: An introduction to Grassland Agriculture by Robert F. Barnes, C. Jerry Nelson, Michael Collins and Kenneth J. Moore.
3. Report of the Sub Group III on Fodder and Pasture Management Constituted under the Working Group on Forestry and Sustainable Natural Resource Management by Planning commission of India.

Title: ASC - 204 Live Stock Management – II

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: The course is designed to impart knowledge about the preparation of livestock accommodation and its maintenance. Also, the placing of livestock in the accommodation and monitoring the livestock placed.

Job Role: The Dairy Farmer/Entrepreneur is expected to take various decisions regarding viability and sustainability of dairy farms. He/she is expected to ensure proper care of dairy animals, their health and productivity, milking and marketing of dairy products.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit – I

Monitoring health and wellbeing of livestock: Promoting health and well-being of animals and minimize stress and injury; Organizational policy in relation to the treatment of animals under welfare legislation; Need for animal movement and how this varies at different times; The appearance, posture and movement, behaviour and bodily functioning of healthy animals characteristics of the species, age, health status and social needs; Signs which indicate potential problems with animals' health and well-being, and the actions to be taken;

Unit – II

Types of preventative care for maintaining the health and well-being of animals; Adjustment of environment for the health and well-being of the animals; Animal temperaments and behaviour related to the associated hazards and risks to animals and staff during animal feeding and cleaning of animal house; Reasons for keeping records and the importance of accuracy; Recognizing a health emergency in an animal and the correct actions to take; Processes that support environment and resource efficiencies; Correct methods for disposing of organic and inorganic waste.

Unit – III

Deliver basic recommended treatment to livestock: Significance of expiry dates on drugs and medication; Possible sources of contamination to medication and how to identify damage; Equipment to select for each treatment; Administering prescribed basic healthcare treatments; Importance of keeping to instructions for basic healthcare treatments; Potential consequences of not keeping to instructions and procedures for carrying out treatments;

Unit – IV

Use of restraint techniques; Reasons and legislative requirements for 'withdrawal periods' for animals; Reasons for personal hygiene and safety; Reasons for maintaining records; Correct methods for disposing of waste; Changes in the condition of the animal which may occur after the treatment; Monitoring the behaviour of animals after treatment andreport unusual signs; Organizational policy in relation to animal treatment and one's responsibility under welfare Legislation

Practical: the trainee is expected to undertake an investigation into the healthcare provision both preventive and curative for the livestock by the dairy farmers and prepare a detailed report.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

1. Handbook of livestock management by Richard A. Battaglia, Pearson Education.
2. Livestock Production and management by N.S.R. Sastry and C.K. Thomas, Kalyani Publishers, Ludhiana, India.
3. Hand book of Animal Husbandry by V.K. Taneja, C. Chakravarty, C.S. Viswanath, Aruna T. Kumar, Indian Council of Agricultural Research, New Delhi.

Title: AGR - 205 Workforce Health & Safety Management

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: to acquaint the trainee about the health and hygiene standards to be maintained by the workforce in the farm. Also it aims to provide the knowledge about various safety measures to be adopted and practiced.

Job Role: The Dairy Farmer/Entrepreneur is expected to take various decisions regarding viability and sustainability of dairy farms. He/she is expected to ensure proper care of dairy animals, their health and productivity, milking and marketing of dairy products.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit – I

Maintenance of Clean & Efficient Workplace: Personal hygiene and fitness requirements in dairy farm; General duties of dairy farm worker under relevant health and safety legislation; Personal protective equipment and clothing required in a dairy farm and how it is cared for;

Unit – II

Safe usage of materials and equipment required in dairy farm; Importance of good housekeeping in dairy farm; Safe disposal methods for livestock waste. Methods for minimizing environmental damage during work

Unit – III

Safety Procedures at dairy farm: Risks to health and safety and the measures to be taken to control those risks in dairy farm; Procedures and requirements for the treatment of injuries/illnesses in dairy farm;

Unit – IV

Basic emergency first aid procedure; Local emergency services; Reporting of accidents, incidents and problems and appropriate action to be taken

Practical: A report is to be prepared about the general health and hygiene standards maintained by the dairy farmer workers and the initiatives being taken for the same by their employers in the dairy farm in the neighbourhood. Comparison of the same with industry standards.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

1. Safety Management: A Comprehensive Approach to Developing a Sustainable System by Chitram Lutchman, Rohanie Maharaj, Waddah Ghanem
2. Occupational Health and Safety Management: A Practical Approach, Third Editionby Charles D. Reese
3. Human Safety and Risk Management: A Psychological Perspective, Third Editionby A. Ian Glendon, Sharon Clarke

Semester III

Paper Title : GC 302 – E-COMMERCE

Theory:80
Internal Assessment: 20
Time:3 hours

Objective : The objective of this paper is to provide fundamental knowledge to the students about E-Commerce so that they can better perform in any area of operation and can excel in the field of commerce with IT specialization.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Electronic Commerce Framework, Basics and Tools of E-Commerce, Comparison of Web-based with Traditional Business; Growth of E-Commerce – Present and potential. E-Business: Meaning, Importance, Models Based on the Relationships of Transacting Parties (B2B, B2C, C2C and C2B), Present Status of E-Commerce in India.

UNIT II

Changing Structure of Organization – The Impact of E-Commerce on supply chain management and customer relation, Socio-Economic Impacts of E-Commerce. Electronic Payment System: Types of Payment System — E-Cash and Currency Servers, E-Cheques, Credit Cards, Smart Cards, Electronic Purses and Debit Cards , Risk and Electronic payment Systems.

UNIT-III

Electronic Data Interchange (EDI): EDI and its applications in business; Legal, Security and Privacy issues in EDI, EDI standardization; Internet based EDI, Internal Information Systems, Work-flow Automation and Coordination, Types of Digital Documents, Digital Signatures.

UNIT-IV

Enterprise Resource Planning: The Emergence of ERP Systems, Business benefits of ERP, ERP Modules and design alternatives, challenges of ERP implementation, Business Process Re-engineering, ERP system development process. Regulatory aspects of E-commerce.

Books Recommended:

- 'Electronic Commerce, A Manager's Guide to E-Business' by Prag Diwan and Sushil Sharma, Vanity Books International, Delhi.
- 'Frontiers of Eletronic Commerce' by Ravi Kalakota, and Andrew B.Shinston, Addision Wesley.
- 'Web Commerce Technology Handbook' by Minoli and Minoli, Tata McGraw Hill, New Delhi.
- 'Electronic Commerce' by Gray P.Schneider, Course Technology, Delhi.
- 'E-Commerce-The Cutting Edge of Business' by Kamlesh Bajaj & Debjani Nag, Tata Mc Graw Hill 2000.
- 'Understanding E-Commerce' by David Kosiur, Microsoft Press. 1997.

Title: ASC 303 Introduction to Poultry Farming

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: the paper aims to provide the trainee with the knowledge of poultry farming as an industry, its role and importance in Indian Economy. The species and sites suitable for poultry farming and their selection.

Job Role: Poultry Farm Assistant is responsible for handling various activities involved in broiler/layer chicken rearing at poultry farm. The individual at work prepares the poultry farm for placement of chicks, carry out their feed; water; litter; brooding and health management to raise the chicks according to market's standards

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

Poultry Industry: an overview of Poultry farming. The importance of poultry products. Socio Economic relevance of poultry farming in India. Poultry development in India: quantitative and qualitative improvement.

Unit II

Role of financial and development agencies in poultry development. Regional concentration of poultry population: state wise growth and concentration of commercial poultry farming.

Unit III

Farming Systems: Free range extensive systems; Backyard extensive systems; Semi Intensive Systems; Intensive System- deep litter, slatted floor and battery cage system.

Unit IV

Selection of farm site & farm building; Poultry equipment. Biology of Poultry: Breeds, Varieties and Strains.

Practical: To carry out a survey to find the regional density of poultry population in immediate neighbourhood also the farming system practiced at large by the poultry farmers. Enumerate the species kept in the in the poultry farms and their cost to a poultry farmer.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

1. Hand book of Applied Broiler Production, Poultry Punch Publication (I), Ltd, New Dehi: Narahari & Kumararaj.
2. Scientific Poultry Production, International Book Distributing Co., Lucknow: P.V. Sreenivasaiah

Title: ASC 304: Management of Broiler & Layer chicken

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: To understand the process of meat & egg production and maximize the returns in rearing of Broiler & Layer chicken.

Job Role: Poultry Farm Assistant is responsible for handling various activities involved in broiler/layer chicken rearing at poultry farm. The individual at work prepares the poultry farm for placement of chicks, carry out their feed; water; litter; brooding and health management to raise the chicks according to market's standards

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit 1

Preparing the brooder house to receive broiler chicks; Brooding & rearing of chicks; Feeding & watering of broilers; Lighting, Ventilation, Litter & Health management

Unit II

Catching & Transportation to processing; Integration in broiler production; Broiler management during different seasons; Broiler production & performance problems

Unit III

Getting started in egg business – Farm location, land requirement, layer buildings for deep litter, cage system & equipment

Unit IV

Management prior to lay (Chick & Grower management) and management during lay (Layer Management) Management practices – Toe, comb & beak trimming. Lighting, Ventilation & Health management, Judging pullets & Factors affecting egg production

Practical: A detailed report to be submitted after assessment of the management of both broiler and layer chicks.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Book Recommended:

1. Hand book of Applied Broiler Production, Poultry Punch Publication (I), Ltd, New Dehi: Narahari & Kumararaj.

2. Scientific Poultry Production, International Book Distributing Co., Lucknow: P.V. Sreenivasaiah
3. Commercial chicken meat and egg production, 5th edition, Spinger : Donald D. Bell, William D. Weaver

Title: ASC 305 Hatchery Management

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: To understand the process of hatchery operations and to produce the good quality chick of broiler & layer.

Job Role: Poultry Farm Assistant is responsible for handling various activities involved in broiler/layer chicken rearing at poultry farm. The individual at work prepares the poultry farm for placement of chicks, carry out their feed; water; litter; brooding and health management to raise the chicks according to market's standards

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

Hatching eggs: Egg Formation & Development of the Embryo; Characters for good quality hatching eggs & Maintaining Hatching egg quality;

Unit II

Handling and care of hatching eggs-selection, cleaning, disinfection, storage and transport; Equipment for hatcheries

Unit III

Incubation & Hatching: Hatchery planning, Design & construction; Principles of incubationOperations in Hatchery – securing, storage, traying/setting of hatching eggs, pre-warming eggs prior to setting, testing of fertility (Candling), pulling the hatch, Grading chicks, holding of chicks & chicks delivery.

Unit IV

Hatchery record management, calculation of fertility & Hatchability of eggs.Factors affecting Fertility & Hatchability. Hatchery sanitation & disposal of hatchery waste.

Practical: Prepare a detailed report on Hatchery Management by visiting a hatchery unit in neighbourhood.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Reference:

1. Hand book of Applied Broiler Production, Poultry Punch Publication (I), Ltd, New Dehi: Narahari & Kumararaj.
2. Scientific Poultry Production, International Book Distributing Co., Lucknow: P.V. Sreenivasaiah
3. Commercial chicken meat and egg production, 5th edition, Spinger : Donald D. Bell, William D. Weaver

Semester IV

Paper Title: GC 402 - PROJECT MANAGEMENT

Theory:80
Internal Assessment: 20
Time:3 hours

Objectives: The objective of this paper is to provide knowledge to students about the essentials of undertaking projects in an organizational environment.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT-I

Concept of Project: Meaning, Characteristics, Classification of Projects, Project Life Cycle and its Phases. Project Manager: Roles and Responsibilities, Project Management as a Profession.

UNIT-II

Generating and Screening Ideas – Steps, Monitoring the Environment, Scouting for Project Ideas, Preliminarily Screening, Project Rating Index. Feasibility Studies – Technical, Financial, Economic, Social, Legal and Managerial.

UNIT-III

Project Appraisal Techniques: Objectives, Types and Methods. Project Risks: Meaning, Types, Measurement of Risk,. Decision Tree Analysis (Basic Concepts only).
Project Evaluation – Meaning, Evaluation v/s Appraisal, Objectives, Types of Evaluation, Techniques.

UNIT-IV

Project Organization and Control – Project Network Analysis (Basic concepts of PERT, CPM, Cost and Time Over Run). Project Reporting: Meaning, Purpose, Process, Requirements of a Good Report, Methods, Principles of Good Reporting System.

Books Recommended:

- Project Management - Choudhary – Tata McGraw Hill Pub.
- Project Management: The Managerial Process (Special Indian Edit.) -Clifford F Gray, Oregon State University.
- Projects : Planning, Analysis, Selection, Financing, implementation and Review -Chandra, Prasanna.

Title: ASC 403 Poultry Nutrition

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: The objective of the course is to impart knowledge about improvement of productivity of meat & egg by scientific feeding of broiler & layer.

Job Role: Poultry Farm Assistant is responsible for handling various activities involved in broiler/layer chicken rearing at poultry farm. The individual at work prepares the poultry farm for placement of chicks, carry out their feed; water; litter; brooding and health management to raise the chicks according to market's standards.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.

- All questions carry equal marks

Unit I

Poultry feeds & feeding: Principles of poultry feeding; Feed ingredients for poultry feeding (Energy & protein feedstuffs); Poultry feed additives & supplements – Vitamins, Minerals, prebiotics & probiotics, enzymes etc.

Unit II

Feeding standards: Feeding standards/specification for broiler & layer; Feed allowance & Dietary requirements; Feed formulation; feeding methods; feed processing & quality control

Unit III

Quality standards: storage of feed, water and maintaining their quality; poultry farm's waste disposal methods; legislation; codes or guidelines of best practice in broilers/layer welfare and accommodation

Unit IV

Balanced feed-feeds for chicks at different stages: age wise scientific feeding programs, types of feed and water requirements for broiler chicks;

Different environmental, feed, water, light requirements to produce quality broilers; documentation of the feed record of the broiler chicks.

Practical: Preparation of report on the feed formulation, management and feeding process. Observations to be made on the quality standards of feed and water prescribed by the industry and the standards maintained.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

1. Hand book of Applied Broiler Production, Poultry Punch Publication (I), Ltd, New Dehi: Narahari & Kumararaj.
2. Scientific Poultry Production, International Book Distributing Co., Lucknow: P.V.Sreenivasaiah
3. Commercial chicken meat and egg production, 5th edition, Spinger : Donald D.Bell, William D. Weaver

Title: ASC404: Poultry Health & Bio security

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: To enable the trainee to understand the poultry immunity system & disease control. To familiarise them with the generally accepted norms of the industry to maintain health.

Job Role: Poultry Farm Assistant is responsible for handling various activities involved in broiler/layer chicken rearing at poultry farm. The individual at work prepares the poultry farm for placement of chicks, carry out their feed; water; litter; brooding and health management to raise the chicks according to market's standards

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

Disease of the chicken; Vaccination – principles & practices; Vaccination schedule for Broiler & Layer; Medication for the prevention & treatment of diseases;

Unit II

Maintenance of vaccination record of the broiler chicks; Policy with respect to purchase of day old chicks from suppliers. Reporting of accidents, incidents and problems and appropriate action to be taken.

Unit III

Role of a broiler farm worker in maintain health standards on the farm. Maintenance of hygiene at poultry farm. Personal protective equipment and clothing required in a poultry farm and how it is cared for; safe usage of materials and equipment required in dairy farm.

Unit IV

Bio-security: Components of Bio security – isolation, traffic control & sanitation; Routine Daily Sanitation Practices; Cleaning & Disinfection of poultry facilities.

Practical: Prepare a detailed report on Health Management i.e. the poultry immunity system & disease control mechanisms and the norms practiced.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

1. Hand book of Applied Broiler Production, Poultry Punch Publication (I), Ltd, New Dehi: Narahari & Kumararaj.
2. Scientific Poultry Production, International Book Distributing Co., Lucknow: P.V. Sreenivasaiah
3. Commercial chicken meat and egg production, 5th edition, Spinger : Donald D. Bell, William D. Weaver

Title: ASC- 405 Poultry Production Economics & Marketing

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: The objective of the course is to understand economics & marketing of broiler & layer farms.

Job Role: Poultry Farm Assistant is responsible for handling various activities involved in broiler/layer chicken rearing at poultry farm. The individual at work prepares the poultry farm for placement of chicks, carry out their feed; water; litter; brooding and health management to raise the chicks according to market's standards

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

Poultry Production Economics: Preparation of Project report for broiler farm & Layer farm; Measures of performance efficiency of poultry; Economics of poultry operation;

Unit II

Marketing: Current marketing system; Market structure; Marketing channel; Constrains of marketing. Maintenance of farm records.

Unit III

Poultry farm expenses: infrastructure, labour, chicks etc. Receipts from poultry enterprises. Profits from poultry enterprises: labour income variation. Feed-Egg/meat ratio: effect on number of eggs and chicken produced.

Unit IV

Factors affecting labour income: size of flock; Egg/meat production. Factors affecting price of egg/chicken: yearly variation; seasonal variation; storage; grade price curve.

Practical: A project report should be prepared on starting a poultry business, infrastructure required, its expected costs and receipts should be estimated by carrying out a field survey.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Semester V

Title : GC 502 - INTRODUCTION TO RESEARCH METHODOLOGY AND REPORT WRITING

Theory:80
Internal Assessment: 20
Time:3 hours

Objective: The objective of the module is to familiarize the students with basics of research methodology like types of research, data collection, sampling techniques and report writing.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Research - Meaning, Characteristics, Types, Process and Utility,. Research Design - Meaning, Types and Features of good research design. Qualitative and Quantitative research. Importance of review of literature. The research proposal.

UNIT II

Primary and Secondary data. Methods of data collection. Measurement and scaling, Designing of Questionnaire and schedule.

UNIT III

Sampling Design and Sampling procedure. Sampling and non sampling errors. Data analysis and interpretation. Role of computers in research.

UNIT IV

Business Reports and Proposals: Introduction, Meaning a Report? Steps in Writing a Routine Business Report, Format of Proposals, Four Key Elements in Winning Business Proposals; Case Studies.

Books Recommended :

- 'Research Methods and Techniques' by C.R. Kothari, Eastern Wiley Publications.
- 'Business Research Methods' by A. Bryman, and E. Bell, Oxford University Press, New York.
- 'Research Methodology A Step by Step Guide for Beginners' by R. Kumar, Pearson Education.

Title: ASC 503 Introduction to Shrimp Farming

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: the paper aims to provide the trainee with the knowledge of shrimp farming as an industry, its role and importance in Indian Economy. The species and sites suitable for shrimp farming and their selection. Also to equip the trainee with the technical knowledge and skills required for designing, construction and management of a pond.

Job Role: The Shrimp farmer is expected to perform the following tasks when at work purchases seeds from hatcheries on deciding the type of species to harvest, constructs a pond and harvest the shrimp by giving it a healthy feed and maintaining quality to sell in the market.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,

- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

Fishery: Role and Importance of fisheries in Indian Economy; Blue Revolution; Export-Import situation of Fisheries. Policy framework for the sector. Shrimp Farming: extent, role and importance in contribution to national income and employment generation. Species of shrimp which can be grown in India: metapenaeusensis, m. monoceros, m. brevicornis, white leg shrimp, penaeusvannamei, penaeussemisulcatus and p. merguiensis. Factors affecting selection the shrimp: seasonal, productivity/yield, salinity, temperature, the food conversion ratios.

Unit II

Factors affecting suitable sites for shrimp farming: appropriate topographical and climatic conditions; transportation; near market (hatcheries and processing plants); electricity; pollution free water (both fresh and brackish); commercial feed; soil quality.

The shrimp farmer's policies on: incentives, delivery standards and personnel management, workflow involved in shrimp farming; importance of the individual's role in the workflow; reporting structure; safety and quality standards followed in the organization

Unit III

Types of culture farming: traditional, semi intensive, intensive farming. Factors influencing the choice of the type of culture farming. Designing the pond: Facilitating water exchange management, product harvesting, collection of waste and elimination and feeding. Prevention of clogging at the inlets. Water: Installation of water pumps; provisioning of a reservoir to store and provide water supply; decision of canal size based on the culture period, efficiency of the pump and the required water exchange rate; preventive measures for floods.

Unit IV

Construction of pond: various shapes and sizes, depth and width, length and breadth of a pond suitable for specific culture. Steps for construction of ponds. Sedimentation pond. Pond Management: Soil sampling; Leaching; Pond drying; Tilling; Control of undesirable species; Liming; Fertilization, Managerial and Construction costs.

Practical: To prepare a detailed report on the species of shrimp being raised and harvested in neighbourhood. The schemes launched by the Centre and State governments to promote shrimp farming in the region. Analysis of topographical, climatic and other factors to decide the type of culture farming and designing of a pond. Techniques adopted by shrimp farmers in the neighborhood to manage ponds.

Instructions for Practical Paper : (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

1. Advances in Shrimp Aquaculture Management by S Felix
2. Aquaculture, Resource Use and the Environment by Boyd, Claude, Mc Nevin and Aaron
3. Marine Shrimp Culture: Principles and Practices edited by A.W. Fast, L.J. Lester
4. Economics of Brackishwater Shrimp Culture by Palparthi Andrew

5. Shrimp culture: pond design, operation and management by P. Kungvankij and T.E. Chua (FAO)

Paper Title:ASC-504 Seed Stocking and Health Management of Shrimps

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: The objective of the course is to equip the students with the knowledge of seed selection, stocking density and appropriate stocking techniques of seeds for shrimp farming. The paper also aims to provide the trainee with the technical knowledge of health management of shrimps. The diseases prevalent and their preventive care methods.

Job Role: The Shrimp farmer is expected to perform the following tasks when at work purchases seeds from hatcheries on deciding the type of species to harvest, constructs a pond and harvest the shrimp by giving it a healthy feed and maintaining quality to sell in the market.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

Selection of seeds for stocking into a pond; Factors impacting the selection of seed for stocking; Selection of Hatchery for purchase of seeds. Importance of seed selection in shrimp farming. Decision on the stocking density based on the system adopted and the species

selected for the culture; Determination of optimum stocking density in a pond (when the farm is ready for operation); Overstocking of seeds and its economic and managerial problems.

Unit II

Appropriate stocking techniques: transportation the seed as per recommended standard procedure; transport the seeds by considering the time of travel as per the specification (day or night time), Prevention of mortality of seed. Adding and nursing of seeds: Estimation of number of seeds and process of adding seeds in the pond. Stocking of seeds in high densities pond; transfer of seeds to another pond; Estimation process of seeds after transfer to another pond.

Unit III

Types of diseases and deficiencies found in different species of shrimp its causes, symptoms, transmission and measure of control. Best farm management practices to decrease the transmission of diseases: efficient pond management; proper quality feed; water quality; disinfection of all tools and utensils used in the process. Drug Administration: Methods for drug administration to shrimp; Precautions to be taken while administering drugs; list of antibiotics banned internationally.

Unit IV

Water Quality Management and health: Bioremediation; Biofilters; Constructed Wetlands and Water Treatment- primary, biological, physico-chemical and disinfectant. Feed management and health. Immunostimulant: Types of Immunostimulant- bacterial, algae derived, animal derived, nutritional factors as Immunostimulant and hormones. Role and importance of Immunostimulant in improving health of shrimps.

Practical:

Preparation of detailed report on seed selection, stocking density, appropriate stocking techniques Addition of seeds to the pond and their nursing for high survival rate during a field trip. Preparing a report on various diseases prevalent and preventive measures taken by shrimp farmers; commonly administered antibiotics; use of Immunostimulant in the area under study.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.

- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

References

1. Advances in Shrimp Aquaculture Management by S Felix
2. Aquaculture, Resource Use and the Environment by Boyd, Claude, Mc Nevin and Aaron
3. Marine Shrimp Culture: Principles and Practices edited by A.W. Fast, L.J. Lester

Paper Title:ASC-505 Feed and Water Management in Shrimp Farming

Theory:40
Practical 40
Internal Assessment: 20
Time:3 hours

Objective: the paper aims to provide the trainee with know-how of the feed and water requirement of shrimp farming and its maintenance as per industry norms.

Job Role: The Shrimp farmer is expected to perform the following tasks when at work purchases seeds from hatcheries on deciding the type of species to harvest, constructs a pond and harvest the shrimp by giving it a healthy feed and maintaining quality to sell in the market.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

Nutrient requirement: type and quality of the raw material used in feed formulation, feed processing; supplementary or complete diet of the shrimp; essential nutrients; provision of required quality and quantity of nutrient requirements (with respect to different species)

Unit II

Feed Storage: storage of feed in a dry, cool and well ventilated place to maintain consistent moisture and temperature; impact of direct sunlight on feed; the expiry of feed; ensuring proper quantity of feed.

Unit III

Feed management: monitoring and maintaining the feed: making feed adjustments according to the mean body weight of the shrimp; cost of feed and decision. Workforce health and hygiene management.

Unit IV

Water Quality: required levels of dissolved oxygen (DO), pH, ammonia, colour of water and water odour in the pond. Management of water quality: use of scientific equipment in order to control the level of dissolved oxygen (DO), pH, ammonia, colour of water and water odour in the pond. Process of addition of necessary nutrients and fertilizers to water.

Practical: To analyze and understand the process of feed formulation, processing and management. To analyze water quality, its nutritional requirement and maintenance. Exposure to scientific equipment in maintenance of water quality and its usage.

Instructions for Practical Paper: (for 40 marks)

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three questions.
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

References:

1. Advances in Shrimp Aquaculture Management by S Felix
2. The Shrimp Book by V AdlaySanz

Semester VI

Paper Title: GC 602 - TOTAL QUALITY MANAGEMENT

Theory:80
Internal Assessment: 20
Time:3 hours

Objective: The objective of the paper is to provide knowledge to students of the concepts of total quality management and to inculcate among them a concern for quality and customer satisfaction.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Introduction: Definition of Quality, Dimensions of Quality, Quality Planning, Quality costs and its Analysis, Basic concepts of Total Quality Management, Principles of TQM.

UNIT II

Quality Council and Statements, Barriers to TQM Implementation. Quality: Customer Perception and Satisfaction, Customer Complaints, Service Quality, Customer Retention, Continuous Process Improvement – Juran Trilogy, PDSA Cycle, 5S, Kaizen, Supplier Partnership –Selection and Rating

UNIT III

Business Process Reengineering (BPR). Benchmarking –Introduction, Reasons, Process, Quality Function Deployment (QFD) – House of Quality, Benefits, Taguchi Quality Loss Function, Total Productive Maintenance (TPM)

UNIT IV

Latest Challenges of Quality, Six Sigma Concepts: application. Quality Systems: Need for ISO 9000 and Other Quality Systems, ISO 9000:2000 Quality System– Elements, Implementation of Quality System, Documentation, Quality Auditing; ISO 14000, Requirements and Benefits.

Books Recommended:

- 'Total Quality Management' by A.V. Feigenbaum, McGraw-Hill Publication.
- 'Total Quality Management' by J.S. Oakland, Butterworth – Hcinemann Ltd., Oxford Press.
- 'Quality Management – Concepts and Tasks' by V. Narayana & N.S.Sreenivasan, New Age International.
- Total Quality Management for Engineers' by Zeiri, Wood Head Publishers.
- 'Total Quality Management' by James R. Evans, Cengage Learning.

Paper Title: ASC 603 - INDUSTRIAL TRAINING

Practical:300

Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce. These credits will be evaluated in Semester VI.

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